Human Following Robot

Using Ultrasonic and IR sensors

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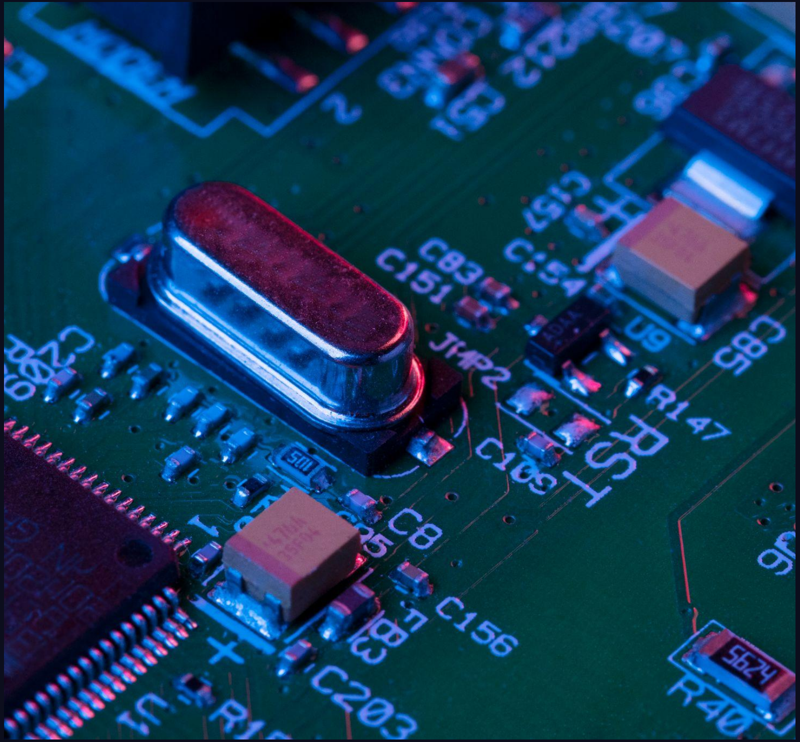
What is our project?

A small autonomous robot that follows a human by detecting distance using **ultrasonic sensors** and **direction using IR sensors**.

Purpose



- Automates human-assistance tasks like carrying luggage or tools
- Reduces the need for manual operation or remote control.

Problems it Solves

Manual carrying effort: People often have to carry heavy luggage or goods themselves.

Inefficiency: Workers in warehouses or hospitals spend extra time moving items.

Accessibility challenges: Elderly or differently-abled individuals may struggle to carry things.

Staff workload: Hospital or retail staff can be overloaded with transporting items.

Real Life Applications

- **Luggage-carrying robots** in airports and hotels that follow passengers.
- **Shopping assistants** that carry items for customers in supermarkets.
- **Warehouse robots** that follow workers to transport goods.
- **Hospital assistants** carrying medicines or equipment for staff.
- **Elderly or differently-abled assistance**: helps carry items while following them.

Achieved Objectives-Progress Report

Basic Features:

- **Human Following using Ultrasonic Sensor:** Robot can detect distance and follow the human based on this distance.

- **Direction Tracking with IR Sensors:** Robot can sense if the human is to the left or right and adjust movement accordingly.

- **Basic Motor Control:** Robot moves forward, turns left/right, and stops based on these sensor inputs.

Achieved Objectives-Progress Report

Extra Features:

- **Pothole/Obstacle Detection:** Robot can detect potholes and stop and reverse to avoid falling/damage.
- **LCD Integration:** Displays the current status of the robot for easier monitoring.
- **Automatic Reversing:** Robot can reverse if it gets too close(less than a specified distance) to the human or obstacle, improving safety.

Cost to implement and run

- Components are **cheap and widely available**: Arduino, sensors, L298N motor driver, small DC motors.
- No cameras or high-end processors → lower cost.
- Maintenance is simple → cheaper to repair or replace parts.

Power Usage

- **Arduino Uno** – ~50 mA
- **Ultrasonic sensor** – ~15 mA
- **IR sensors** – ~10–20 mA
- **Motors** – the biggest power consumers, ~300–500 mA depending on load

Total: ~1A at 5–7V

(Currently powered via DC supply for testing; in real-life, a portable battery can replace it)

Efficiency Notes

- Most power is consumed by motors, so reducing unnecessary movement saves battery.
- Sensors consume very little, making the system energy-efficient overall.
- Using low-power components keeps operation cost low and battery life long.

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For Optimization:

- Switch to brushless motors for higher efficiency and longer lifespan.
- Use sleep mode for Arduino or sensors when idle.
- Optimize motor speed and movement path to minimize energy waste.

Limitations and Real-Life Solutions

Limitation	Current Robot	How Real Robots Solve it
IR sensor affected by sunlight	Stops working properly in bright light	Use advanced IR modules or cameras with image processing
Ultrasonic sensor accuracy	Angled surfaces may give wrong distance	Use LiDAR (uses laser pulses) or multiple sensors
Motor precision	Turns may not be perfectly smooth	Use brushless motors + encoders for exact movement
Navigation in crowded areas	Can’t follow human through obstacles	Use AI-based vision tracking and obstacle mapping

Contribution of each member

Member 1: Ardra P
Sensor Integration & Wiring

- Set up and wired ultrasonic and IR sensors.
- Tested and fixed sensor readings for accurate detection.
- Debugging sensor errors.

Contribution of each member

Member 2: Sayana Sajan
Arduino Programming & Algorithm

- Developed the main Arduino code for human following.
- Implemented motor control logic.
- Added extra features like reversing when too close and LCD display updates.

Contribution of each member

Member 3: Adithya Krishna
Chassis, Motors & System Testing

- Designed and assembled the robot chassis.
- Connected and configured the L298N motor driver to control both motors .
- Integrated extra features like reversing and pothole detection.